

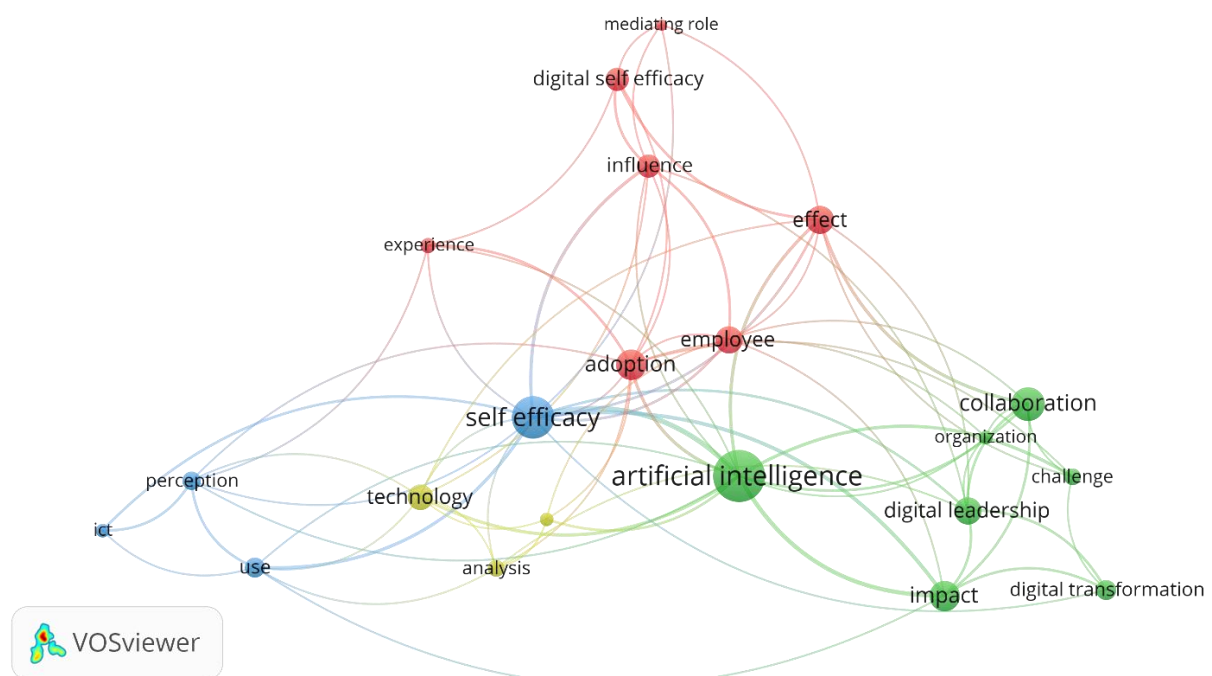
Bibliometric analysis

Bibliometric analysis was used to examine how key terms in the literature appear together. Relevant research articles were retrieved using Publish or Perish. The keyword search was conducted using the study's variables, the publication period was restricted to 2016-2026, and 200 relevant publications were obtained and exported in CSV format for term mapping.

The CSV file was then uploaded into VOSviewer to build a term co-occurrence map. A map based on text data was created by selecting the title field from the bibliographic dataset. Binary counting was applied. The minimum number of occurrences of a term was adjusted according to the threshold (15-25 terms taken), and 21 terms were identified and included in the bibliometric mapping.

A bibliometric visualization showing the relationships between the terms were generated. After generating the term map, the selected terms and their document-term relationships were exported from VOSviewer as two separate files (terms and term relations file).

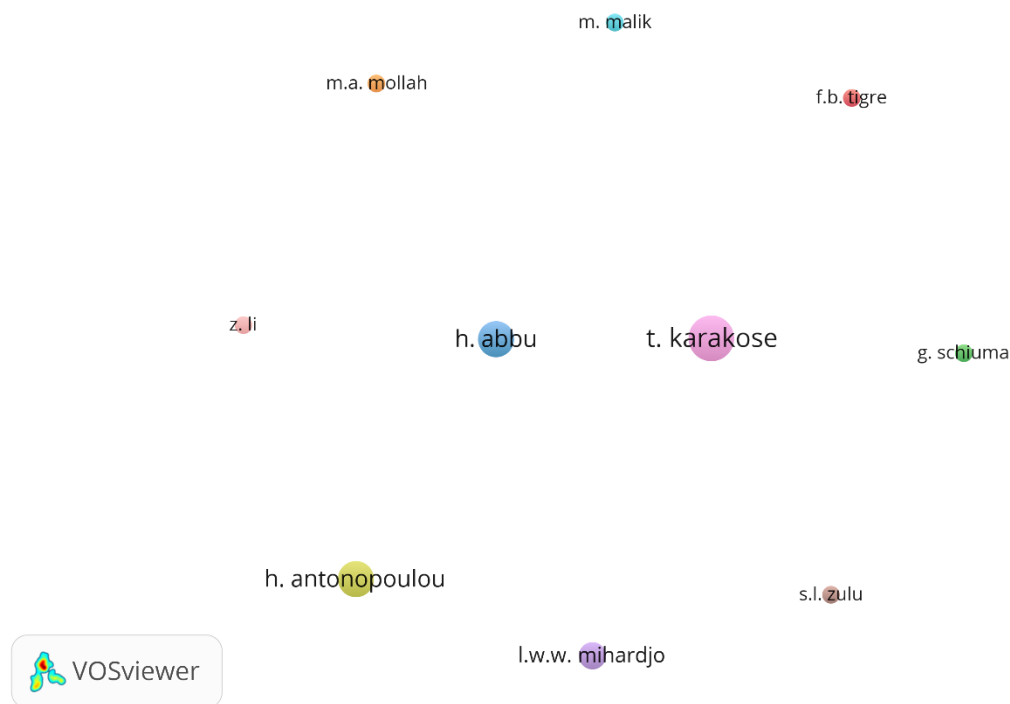
Keyword co-occurrence analysis



Based on the bibliometric network shown in the figure, the keywords related to Digital Leadership, Digital Self-Efficacy, Employer Preparedness for ICT and AI Support appear

within the same bibliometric map and across multiple clusters. The visualization indicates that these constructs are linked through shared connections and co-occurrences, though they form separate thematic clusters. Simply put, overall, the map suggests that the constructs are related within the broader digital transformation literature, with some overlap in research focus, but the strength of connections varies across themes.

Author Co-authorship Analysis



The CSV file retrieved from Pulish or Perish was imported into VOSviewer, where a minimum threshold of two documents per author was applied, resulting in a network of ten authors meeting the eligibility criteria and were retained for network visualization.

Two nodes appear marginally larger than the rest, reflecting a slightly higher volume of relevant publications from those authors. No connecting lines is seen between any of the nodes, indicating a complete absence of co-authorship relationships among the identified scholars that is scholarly contributions across these research themes have emerged largely from independent research efforts rather than collaborative networks.

Factor Analysis

The exported files from VOS viewer, were processed in Microsoft Excel to create a document–term matrix. From the document-term relations file, the document ID and term ID fields were used to create a pivot table, with documents as rows and terms as columns. The matrix was cleaned by replacing zero values and retaining only numerical values. Correlation analysis was then performed using the Data Analysis ToolPak in Excel and the correlation matrix was transposed for factor analysis, which was performed using SPSS. To assess the suitability of the data for factor analysis, Kaiser–Meyer–Olkin (KMO) and Bartlett’s Test were examined. Factor extraction was followed by Varimax rotation. The Rotated Component Matrix was exported to Excel. Only the highest loading value in each row of the rotated component matrix was retained. Each factor was treated as a cluster of related terms.

Rotated component matrix

Rotated Component Matrix ^a									
Component	1	2	3	4	5	6	7	8	9
Adoption				0.751					
Analysis			0.732						
Artificial Intelligence							0.367		
Challenge							0.100		
Collaboration					0.085				
Digital Leadership					0.719				
Digital self efficacy		0.721							
Digital transformation					0.666				
Effect		0.384							
Employee									0.102
Experience				0.795					
Human Resource Management			0.644						
ICT	0.823								
Impact								0.322	
Influence		0.585							

Mediating role		0.700							
Organization							0.106		
Perception	0.851								
Self efficacy						0.491			
Technology			0.632						
Use	0.639								

Through this procedure, 9 clusters were formed. The terms belonging to each cluster were grouped accordingly. This grouping constituted the content analysis.

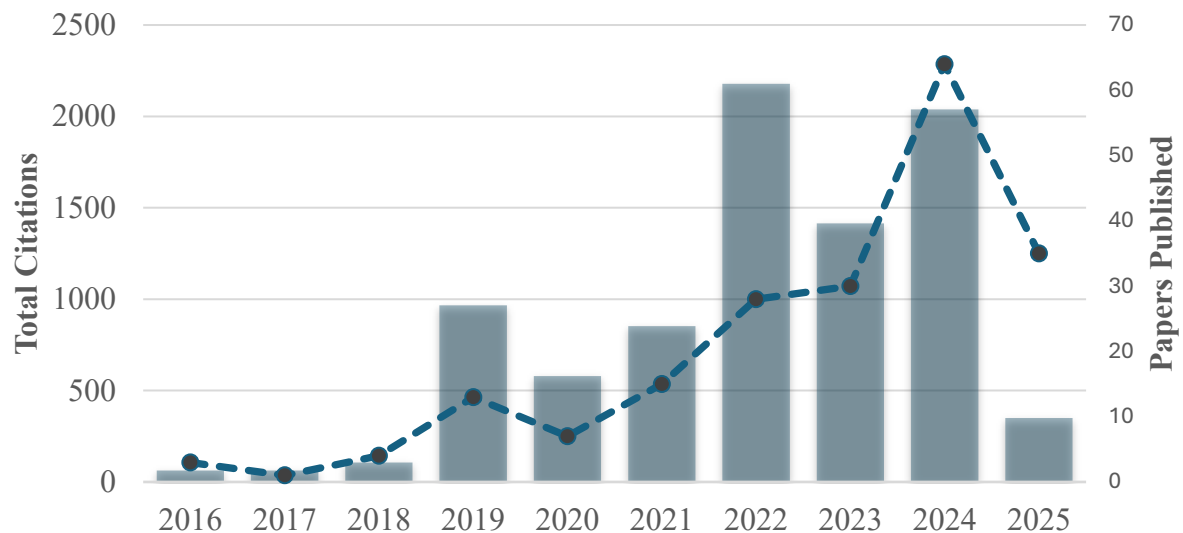
Cluster table

Cluster 1	ICT, Perception, Use
Cluster 2	Digital self efficacy, Effect, Influence, Mediating role
Cluster 3	Analysis, Human Resource Management, Technology
Cluster 4	Adoption, Experience
Cluster 5	Collaboration, Digital Leadership, Digital transformation
Cluster 6	Self efficacy
Cluster 7	Artificial Intelligence, Challenge, Organisation,
Cluster 8	Impact
Cluster 9	Employee

The analysis shows that the research is broadly grouped around technology, people, and organizations. Basically, some clusters focus on the use and adoption of digital technologies, while others discuss employees' abilities and confidence in using them. There is also attention on leadership, collaboration, and digital transformation, highlighting how organizations manage change. A few clusters reflect new technologies like artificial intelligence and their challenges. Overall, the clusters suggest that existing research broadly examines employees, technology adoption, leadership, and organizational impact within the digital work environment.

Citation Analysis

Publication Trend and Year-wise Citations



The publication trend analysis was carried out using Scopus-indexed data covering the period from 2016 to 2025, tracking both annual paper output and total citations accumulated across the relevant literature. Honestly, the dataset comprises 200 publications generating a grand total of 8,609 citations across the decade. Research activity in this area was minimal in the early years, with only 3 papers and 62 citations recorded in 2016 and similarly low figures through 2018. A notable shift occurred from 2019 onward, with output and citations rising steadily 2022 marked a big spike with 28 papers and 2,178 citations, the highest citation count in the dataset. As it turns out, growth continued strongly through 2023 and 2024, with 64 papers published in 2024 alone representing the peak in research volume. Honestly, the 2025 figures reflect a partial year at the time of extraction and are not indicative of a declining trend. Overall, the data confirms that scholarly interest in the constructs examined by this study has grown substantially and consistently over the past decade, establishing a solid and expanding literature base.

Table 1.3: Top 10 Most Cited Works:

Title	Total Cites
Artificial intelligence, robotics, advanced technologies and human resource management: a systematic review	759
Unlocking the value of artificial intelligence in human resource management through AI capability framework	617
Industry 5.0 and Human-Robot Co-working	599
Service robots, customers and service employees: what can we learn from the academic literature and where are the gaps?	472
Preparing Workplaces for Digital Transformation: An Integrative Review and Framework of Multi-Level Factors	378
When and how artificial intelligence augments employee creativity	341
AI-employee collaboration and business performance: Integrating knowledge-based view, socio-technical systems and organisational socialisation framework	299
A multilevel review of artificial intelligence in organizations: Implications for organizational behavior research and practice	284
Artificial intelligence and human workers interaction at team level: a conceptual assessment of the challenges and potential HRM strategies	226
The rise of artificial intelligence – understanding the AI identity threat at the workplace	216

The ten most cited works in the dataset span themes of AI in human resource management, human-robot collaboration, digital transformation readiness, and AI-employee interaction all directly relevant to the constructs examined in this study. The most cited work, a systematic review on artificial intelligence, robotics, and HRM accumulated 759 citations, reflecting its broad foundational influence across the literature. Works examining AI capability frameworks in HRM, Industry 5.0 and human-robot co-working, and digital transformation readiness collectively anchor the AI support and employer ICT preparedness dimensions of the study. The presence of studies addressing AI augmentation of creativity, AI-employee collaboration and business performance, and human-AI interaction strategies among the top ten confirms that the human and organizational dimensions of AI integration have become central concerns in contemporary management research, directly validating the theoretical relevance of the present study's conceptual model.

CONCEPTUAL FRAMEWORK

